

COMPUTER NETWORK LAB

Experiment No. 1

Title

Straight-Through and Cross-Over Cable Configuration

Aim

To prepare and test Straight-Through and Cross-Over Ethernet cables using UTP cable and RJ-45 connectors for network communication.

Apparatus Required

1. UTP (Unshielded Twisted Pair) Cable (Cat5e/Cat6)
2. RJ-45 Connectors
3. Crimping Tool
4. Cable Tester
5. Wire Stripper/Cutter
6. Two Computers or Network Devices
7. Switch/Hub (for testing Straight-Through cable)

Theory

Ethernet cables are used to connect devices in a computer network. The two commonly used cable configurations are:

Straight-Through Cable

- Used to connect different types of devices.
- Example: PC to Switch, Router to Switch.
- Both ends follow the same wiring standard (T568A–T568A or T568B–T568B).

Cross-Over Cable

- Used to connect similar devices directly.
- Example: PC to PC, Switch to Switch.
- One end follows T568A and the other end follows T568B.

Color Coding Standards

T568A Standard

1. White-Green
2. Green
3. White-Orange
4. Blue
5. White-Blue
6. Orange

7. White-Brown
8. Brown

T568B Standard

1. White-Orange
2. Orange
3. White-Green
4. Blue
5. White-Blue
6. Green
7. White-Brown
8. Brown

Procedure

Part A: Straight-Through Cable

1. Cut the UTP cable to the required length.
2. Strip approximately 2 cm of the outer insulation from both ends.
3. Untwist and arrange the wires according to the T568B standard.
4. Trim the wires evenly.
5. Insert the wires into an RJ-45 connector.
6. Use the crimping tool to secure the connector.
7. Repeat the same T568B arrangement on the other end.
8. Test the cable using a cable tester.
9. Connect a PC to a switch and verify network connectivity.

Part B: Cross-Over Cable

1. Prepare one end using the T568A standard.
2. Prepare the other end using the T568B standard.
3. Crimp RJ-45 connectors on both ends.
4. Test the cable using a cable tester.
5. Connect two PCs directly using the cable.
6. Assign IP addresses and verify communication using the ping command.

Observation

Straight-Through Cable

Pin Number	End A (T568B)	End B (T568B)
1	White-Orange	White-Orange
2	Orange	Orange
3	White-Green	White-Green
4	Blue	Blue
5	White-Blue	White-Blue
6	Green	Green
7	White-Brown	White-Brown
8	Brown	Brown

Cross-Over Cable

Pin Number	End A (T568A)	End B (T568B)
1	White-Green	White-Orange
2	Green	Orange
3	White-Orange	White-Green
4	Blue	Blue
5	White-Blue	White-Blue
6	Orange	Green
7	White-Brown	White-Brown
8	Brown	Brown

Result

Successfully prepared and tested Straight-Through and Cross-Over Ethernet cables. Network connectivity was verified through cable testing and successful communication between connected devices.

Experiment No. 2

Title

IP Address Configuration and Network Testing

Aim

To configure IP addresses on computers and verify network connectivity using network testing commands.

Apparatus Required

1. Two or more Computers
2. Switch/Hub
3. Straight-Through Ethernet Cables
4. Network Interface Cards (NICs)
5. Windows/Linux Operating System

Theory

An IP (Internet Protocol) address is a unique logical address assigned to a device on a network. Devices within the same network must have unique IP addresses and the same subnet mask to communicate directly.

Common network testing utilities include:

- **ipconfig** (Windows): Displays IP configuration details.
- **ping**: Tests connectivity between two devices.
- **tracert/traceroute**: Displays the route taken by packets.
- **arp**: Displays and manages the Address Resolution Protocol cache.

Network Configuration

PC-1

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC-2

- IP Address: 192.168.1.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

Procedure

Part A: Configure IP Address

1. Connect the computers using a switch and Ethernet cables.
2. Open **Network and Sharing Center**.
3. Select **Change Adapter Settings**.
4. Right-click **Ethernet** and select **Properties**.
5. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**.
6. Choose **Use the following IP address**.
7. Enter the IP Address, Subnet Mask, and Default Gateway.
8. Click **OK** and save the settings.
9. Repeat the process for the second computer using a different IP address.

Part B: Verify IP Configuration

1. Open Command Prompt.
2. Type:

```
ipconfig
```

3. Verify that the configured IP address and subnet mask are displayed correctly.

Part C: Test Network Connectivity

1. Open Command Prompt on PC-1.
2. Execute:

```
ping 192.168.1.20
```

3. Observe the replies received from PC-2.
4. Open Command Prompt on PC-2.
5. Execute:

```
ping 192.168.1.10
```

6. Verify successful communication.

Part D: Display Routing Information

1. Execute:

```
tracert 192.168.1.20
```

2. Observe the route followed by packets.

Observations

IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC-1	192.168.1.10	255.255.255.0	192.168.1.1
PC-2	192.168.1.20	255.255.255.0	192.168.1.1

Ping Test Result

Source	Destination	Status
PC-1	PC-2	Successful
PC-2	PC-1	Successful

Result

Successfully configured IP addresses on networked computers and verified network connectivity using the **ipconfig**, **ping**, and **tracert** commands. Communication between devices was established successfully.

Experiment No. 3

Title

Network Device Configuration (Hub, Switch, and Router)

Aim

To study and configure basic network devices such as Hub, Switch, and Router and understand their role in a computer network.

Apparatus Required

1. Hub
2. Switch
3. Router
4. Two or More Computers
5. UTP Ethernet Cables
6. Cisco Packet Tracer (Optional)
7. Console Cable (for Router Configuration)

Theory

Hub

A Hub is a Layer 1 (Physical Layer) networking device that connects multiple computers in a network. It broadcasts incoming data to all connected devices without filtering.

Switch

A Switch is a Layer 2 (Data Link Layer) device that forwards data only to the intended destination using MAC addresses. It reduces network traffic and collisions.

Router

A Router is a Layer 3 (Network Layer) device that connects different networks and forwards packets based on IP addresses. It determines the best path for data transmission.

Procedure

Part A: Hub Configuration and Testing

1. Connect two or more computers to the Hub using Ethernet cables.
2. Configure IP addresses on all computers within the same network.

Example:

Computer	IP Address	Subnet Mask
PC-1	192.168.1.10	255.255.255.0
PC-2	192.168.1.20	255.255.255.0

3. Open Command Prompt.
4. Verify connectivity using:

```
ping 192.168.1.20
```

5. Observe successful communication between computers.

Part B: Switch Configuration and Testing

1. Connect all computers to the Switch.
2. Assign IP addresses within the same subnet.
3. Verify physical connectivity through switch port LEDs.
4. Test communication using the ping command.
5. Observe improved performance compared to a Hub.

Part C: Router Configuration

1. Connect the Router to the computer using a console cable.
2. Open terminal software (PuTTY/Tera Term).
3. Access Router CLI.
4. Enter privileged mode:

```
Router> enable
```

5. Enter global configuration mode:

```
Router# configure terminal
```

6. Configure the router interface:

```
Router(config)# interface gigabitethernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
```

7. Exit configuration mode:

```
Router(config-if)# exit
Router(config)# exit
```

8. Save configuration:

```
Router# write memory
```

9. Configure PCs with the router IP address as the default gateway.
10. Verify connectivity between the PCs and the router using:

```
ping 192.168.1.1
```

Result

Successfully studied and configured Hub, Switch, and Router devices. Network connectivity was verified using the ping command, and router interface configuration was completed successfully.

Experiment No. 4

Title

LAN Setup Using Different Network Topologies

Aim

To set up a Local Area Network (LAN) using different network topologies and study their operation and performance.

Apparatus Required

1. Computers (4 or more)
2. Hub/Switch
3. Router (optional)
4. UTP Ethernet Cables
5. RJ-45 Connectors
6. Crimping Tool
7. Network Interface Cards (NICs)
8. Cisco Packet Tracer (optional)

Theory

A Network Topology refers to the physical or logical arrangement of devices in a network. Different topologies provide different levels of performance, reliability, and scalability.

Star Topology

- All devices are connected to a central switch or hub.
- Easy to install and manage.
- Failure of one cable does not affect the entire network.

Bus Topology

- All devices share a common communication cable.
- Simple and inexpensive.
- Failure of the backbone cable affects the entire network.

Ring Topology

- Devices are connected in a circular manner.
- Data travels from one node to another around the ring.
- Failure of one node may affect communication.

Mesh Topology

- Every device is connected to every other device.
- Provides high reliability and fault tolerance.
- Requires a large number of connections.

Procedure

Part A: Star Topology Setup

1. Connect all computers to a central switch using straight-through cables.
2. Assign IP addresses in the same subnet.

Computer	IP Address
----------	------------

PC-1	192.168.1.10
PC-2	192.168.1.11
PC-3	192.168.1.12
PC-4	192.168.1.13

3. Verify connectivity using the ping command.
4. Record the results.

Part B: Bus Topology Simulation

1. Create a Bus topology using network simulation software such as Cisco Packet Tracer.
2. Assign IP addresses to all nodes.
3. Send packets between devices.
4. Observe data transmission.

Part C: Ring Topology Simulation

1. Connect devices in a ring configuration using Packet Tracer.
2. Configure IP addresses.
3. Test communication between nodes.
4. Observe packet flow around the ring.

Part D: Mesh Topology Simulation

1. Connect each device with every other device.
2. Configure IP addresses.
3. Verify communication between all devices.
4. Observe fault tolerance characteristics.

Observations

Star Topology

Source	Destination	Result
PC-1	PC-2	Successful
PC-1	PC-3	Successful
PC-1	PC-4	Successful

Bus Topology

Test	Result
Data Transmission	Successful
Backbone Failure	Network Failure

Ring Topology

Test	Result
Data Transmission	Successful
Node Failure	Communication Affected

Mesh Topology

Test	Result
Data Transmission	Successful
Link Failure	Alternate Path Available

Result

Successfully set up and studied a Local Area Network (LAN) using different network topologies namely Star, Bus, Ring, and Mesh. Connectivity between devices was verified and the characteristics of each topology were observed.

Experiment No. 5

Title

Router Configuration Using Cisco Packet Tracer

Aim

To configure a Cisco Router using Cisco Packet Tracer and establish communication between two different networks.

Apparatus Required

1. Computer with Cisco Packet Tracer installed
2. One Cisco Router (e.g., 2911 Router)
3. Two Switches
4. Four PCs
5. Copper Straight-Through Cables

Theory

A Router is a Layer 3 (Network Layer) device used to connect different networks. It forwards packets based on IP addresses and routing tables. Cisco Packet Tracer provides a simulation environment for configuring and testing routers without physical hardware.

Network Diagram

Network 1:

- PC0: 192.168.1.10/24
- PC1: 192.168.1.11/24
- Gateway: 192.168.1.1

Network 2:

- PC2: 192.168.2.10/24
- PC3: 192.168.2.11/24
- Gateway: 192.168.2.1

Router Interfaces:

- GigabitEthernet0/0 → 192.168.1.1/24
- GigabitEthernet0/1 → 192.168.2.1/24

Procedure

Step 1: Create Network Topology

1. Open Cisco Packet Tracer.
2. Place one Router, two Switches, and four PCs in the workspace.
3. Connect:
 - PC0 and PC1 to Switch0.
 - PC2 and PC3 to Switch1.
 - Switch0 to Router GigabitEthernet0/0.
 - Switch1 to Router GigabitEthernet0/1.

Step 2: Configure IP Addresses on PCs

PC0

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC1

- IP Address: 192.168.1.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC2

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

PC3

- IP Address: 192.168.2.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

Step 3: Configure Router Interfaces

Open Router CLI and enter the following commands:

```
Router> enable
Router# configure terminal

Router(config)# interface gigabitethernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface gigabitethernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# exit
Router# write memory
```

Step 4: Verify Router Configuration

Display interface information:

```
Router# show ip interface brief
```

Expected Output:

Interface	IP Address	Status
GigabitEthernet0/0	192.168.1.1	up
GigabitEthernet0/1	192.168.2.1	up

Step 5: Test Connectivity

From PC0:

```
ping 192.168.2.10
```

From PC2:

```
ping 192.168.1.10
```

Result

Successfully configured a Cisco Router in Cisco Packet Tracer and established communication between two different IP networks. The configured router forwarded packets correctly between the networks.

Experiment No. 6

Title

Static and Dynamic Routing Using Cisco Packet Tracer

Aim

To configure and verify Static Routing and Dynamic Routing between multiple networks using Cisco Packet Tracer.

Apparatus Required

1. Computer with Cisco Packet Tracer installed
2. Three Cisco Routers
3. Three Switches
4. Three PCs
5. Copper Straight-Through Cables

Theory

Static Routing

Static Routing is a routing technique in which routes are manually configured by the network administrator. It is suitable for small networks and provides greater control over routing paths.

Dynamic Routing

Dynamic Routing automatically learns and updates routes using routing protocols such as RIP, OSPF, and EIGRP.

Network Topology

Device	Interface	IP Address
Router0	G0/0	192.168.1.1/24
Router0	G0/1	10.0.0.1/30
Router1	G0/0	10.0.0.2/30
Router1	G0/1	10.0.0.5/30
Router2	G0/0	10.0.0.6/30
Router2	G0/1	192.168.3.1/24
PC0	NIC	192.168.1.10/24
PC2	NIC	192.168.3.10/24

Procedure

Part A: Configure Router Interfaces

Router0

```
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface g0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
```

Router1

```
enable
configure terminal

interface g0/0
ip address 10.0.0.2 255.255.255.252
no shutdown

interface g0/1
ip address 10.0.0.5 255.255.255.252
no shutdown
```

Router2

```
enable
configure terminal

interface g0/0
ip address 10.0.0.6 255.255.255.252
no shutdown

interface g0/1
ip address 192.168.3.1 255.255.255.0
no shutdown
```

Part B: Static Routing Configuration

Router0

```
ip route 192.168.3.0 255.255.255.0 10.0.0.2
```

Router1

```
ip route 192.168.1.0 255.255.255.0 10.0.0.1
ip route 192.168.3.0 255.255.255.0 10.0.0.6
```

Router2

```
ip route 192.168.1.0 255.255.255.0 10.0.0.5
```

Verify Static Routes

```
show ip route
```

Test Connectivity

From PC0:

```
ping 192.168.3.10
```

Successful replies indicate proper routing.

Part C: Dynamic Routing Configuration (RIP Version 2)

Remove Static Routes (Optional)

```
no ip route network-address subnet-mask next-hop
```

Router0

```
router rip
version 2
network 192.168.1.0
network 10.0.0.0
no auto-summary
```

Router1

```
router rip
version 2
network 10.0.0.0
no auto-summary
```

Router2

```
router rip
version 2
network 192.168.3.0
network 10.0.0.0
no auto-summary
```

Verify RIP Routes

```
show ip route
```

Routes learned through RIP are marked with the letter **R**.

Test Connectivity

```
ping 192.168.3.10
```

Result

Successfully configured and verified both Static Routing and Dynamic Routing (RIP Version 2) using Cisco Packet Tracer. Communication between different networks was established successfully through both routing methods.

Experiment No. 7

Title

DNS and DHCP Configuration Using Cisco Packet Tracer

Aim

To configure DNS (Domain Name System) and DHCP (Dynamic Host Configuration Protocol) services and verify automatic IP address assignment and domain name resolution.

Apparatus Required

1. Computer with Cisco Packet Tracer installed
2. One Router
3. One Switch
4. Two PCs
5. One Server
6. Copper Straight-Through Cables

Theory

DHCP (Dynamic Host Configuration Protocol)

DHCP is a network protocol used to automatically assign IP addresses, subnet masks, default gateways, and DNS server addresses to client devices.

Advantages:

- Automatic IP configuration
- Reduces administrative effort
- Prevents IP address conflicts

DNS (Domain Name System)

DNS translates human-readable domain names (e.g., www.example.com) into IP addresses that computers use for communication.

Network Topology

Device	IP Address
Router	192.168.1.1
Server (DNS & DHCP)	192.168.1.2
PC0	DHCP Assigned
PC1	DHCP Assigned

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.2

Procedure

Part A: Create the Network

1. Open Cisco Packet Tracer.
2. Add one Router, one Switch, one Server, and two PCs.
3. Connect all devices using straight-through cables.
4. Configure the Router interface:

```
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

exit
write memory
```

Part B: Configure DHCP Server

Server Configuration

1. Click on the Server.
2. Select **Desktop** → **IP Configuration**.
3. Assign:

Parameter	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

4. Open **Services** → **DHCP**.
5. Turn DHCP Service **ON**.

Configure DHCP Pool:

Parameter	Value
Pool Name	LAN_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.100
Subnet Mask	255.255.255.0
Maximum Users	50

6. Click **Add**.

Part C: Configure DNS Server

1. Select **Services** → **DNS**.
2. Turn DNS Service **ON**.
3. Add a DNS record:

Name	Address
<u>www.college.com</u>	192.168.1.2

4. Click **Add**.

Part D: Configure Client PCs

PC0

1. Desktop → IP Configuration.
2. Select **DHCP**.

PC1

1. Desktop → IP Configuration.
2. Select **DHCP**.

Observe that IP addresses are assigned automatically.

Example:

Device	Assigned IP
PC0	192.168.1.100
PC1	192.168.1.101

Part E: Verification

Verify DHCP Assignment

On PC0:

```
ipconfig
```

Check the assigned IP address, gateway, and DNS server.

Test Connectivity

```
ping 192.168.1.2
```

Successful replies indicate proper connectivity.

Verify DNS Resolution

Open Web Browser on PC0 and enter:

`www.college.com`

The DNS server should resolve the domain name to the server IP address.

Result

Successfully configured DHCP and DNS services using Cisco Packet Tracer. Client PCs obtained IP addresses automatically through DHCP, and domain names were resolved successfully using the DNS server.